D03 · Shop Floor Execution & MES

05 · Andon, escalation & real-time response

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1 WHAT IT IS

Andon is the signal-to-action loop on the floor — the visual, auditory, or system-driven alert that tells the right person there is a problem that needs attention, plus the escalation matrix that determines how long until the next level gets involved. It is the operational equivalent of an alarm system: only as useful as the response discipline behind it.

2 WHY IT MATTERS

Andon is one of the foundational Toyota Production System pillars and one of the most consistently mis-implemented **(Liker, 2004 — TPS scholarship)**. **The system tells operators they can stop the line; the culture tells them they cannot**. The disconnect is the failure mode: andon hardware deployed without the response discipline becomes decoration. The economic case for andon is well-established: a problem caught and resolved within the first 5 minutes of occurrence costs roughly an order of magnitude less than the same problem caught at end-of-shift, and two orders of magnitude less than the problem caught by the customer. **The asymmetry between resolution cost and detection delay is exponential, not linear**, which is what makes andon economically catastrophic to skip.

3 HOW IT IS DONE

1. **Define the andon trigger criteria by issue type — quality, equipment, material, safety.** A single "something is wrong" signal is useless; the response is different for each issue class. The trigger criteria must be specific enough that the operator does not hesitate (call it on a quality drift over X, equipment alarm of severity Y, material shortage that will hit zero in < Z minutes).
2. **Build the escalation matrix with explicit time-to-response by level.** Level 1 (team leader) within 60 seconds; Level 2 (shift supervisor) within 5 minutes; Level 3 (plant manager) within 15 minutes. The matrix is published, visible at every cell, and the breaches are tracked as a KPI.
3. **Make the andon physically and digitally inescapable.** Light tower at the cell, audible signal at zone level, push notification to the responder's device, plus a digital escalation timer that triggers the next level if response isn't acknowledged. The signal cannot be missed and cannot be silenced without acknowledgment.
4. **Protect the operator who pulls the andon — explicitly and visibly.** If the operator who calls an andon is later questioned about "why did you stop the line," the next andon never gets pulled. Leadership must visibly thank, not interrogate. This is the cultural piece that determines whether the hardware delivers value or sits idle.
5. **Treat every andon as a learning event, not a problem to be cleared.** Each andon should generate a brief root cause record and feed into the kaizen pipeline. An andon that recurs daily without root cause work is a symptom of a system that resolves but does not learn.
6. **Track andon-response KPIs as leading indicators of operational health.** Time-to-acknowledge, time-to-resolve, andon recurrence rate, and the ratio of acknowledged-but-unresolved at end-of-shift. These metrics are leading indicators; OEE and defect rates are lagging indicators of the same underlying reality.

WORKED EXAMPLE — ILLUSTRATIVE

An aerospace machining facility deploys andon at 28 cells with a tiered escalation matrix (60s / 5min / 15min). Baseline: andon pulled an average of 4 times per shift, with median resolution time of 47 minutes. The implementation focuses on three things: physical light towers + digital notifications, leadership protection of operators who pull, and a daily 15-minute andon review meeting with root cause records. At 6 months: **andon frequency rises to 22 per shift** (operators trust the system now), **median resolution time drops to 9 minutes**, and the recurring andon types fall by 67% as root cause work eliminates them. OEE rises 8.7 points; quality escapes to customers drop 42%. The counter-intuitive lesson is that the number of andons pulled is not the metric to minimize — the metric is the time to resolution and the recurrence rate.

IN PRACTICE

Leadership protects the operator who pulls — publicly

Explicit leadership statements that pulling the andon is the operator's job, not their right **(Liker, 2004 — TPS scholarship)**. Public recognition for pulls that prevented defects. Zero tolerance for the supervisor who asks "why did you stop?"

Track response KPIs at the same visibility as OEE

Time-to-acknowledge and time-to-resolve published at the same level as OEE, reviewed monthly with the same attention. Without that, andon decays into a dashboard nobody watches.

Tight first-level trigger criteria — avoid alert fatigue

If team leaders are pulled into too many andons because criteria are too loose, they stop responding promptly. The escalation matrix must be calibrated to keep first-level response rate above 95% — anything below that signals trigger thresholds need tightening.

End-of-shift accountability for unresolved andons

Without explicit accountability at shift end, the andon system becomes a queue rather than a response loop. The shift report includes andons opened, andons closed, and andons handed over with named owners — every shift, no exceptions.

WHAT YOU WOULD USE

ANDON PLATFORMS

MES-integrated andon (Siemens, Rockwell, AVEVA) — When the MES is already in place, native andon modules give the tightest integration with work orders and OEE, with escalation rules configured in the same place **(Siemens — vendor; Rockwell — vendor)**.

Standalone andon platforms (Andonix, ProShop, Mevisio) — Specialized systems that integrate light towers, digital notifications, and escalation logic — faster to deploy than full MES, often the right entry point **(Andonix — vendor)**.

Mobile-first andon (Tulip, custom apps) — Andon delivered through tablets and phones with push notifications and digital escalation timers — strong fit for distributed or mobile workforces **(Tulip — vendor)**.

Physical light tower + Slack/Teams notification — The lean DIY combination — light tower triggered by PLC, escalation via existing messaging platform. Lower ceiling but rapid deployment; suits operations that are still building the andon culture.

4 SO WHAT

> **Andon hardware without response discipline is decoration; response discipline without leadership protection of the operator who pulls is theater.** Define triggers by issue type, build the escalation matrix with explicit response times, make the signal inescapable, protect the operator publicly, treat every andon as a learning event, and track response KPIs as leading indicators. The plants that solve this catch problems in minutes rather than shifts — and the difference compounds into every other operational metric.

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